

Capture The Flag (CTF) Machine

Report

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Class : MCA – A (2nd Semster)

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GitHub Repository Link: <https://github.com/sjgowshicksidharth-rgb/CTF-Machine>

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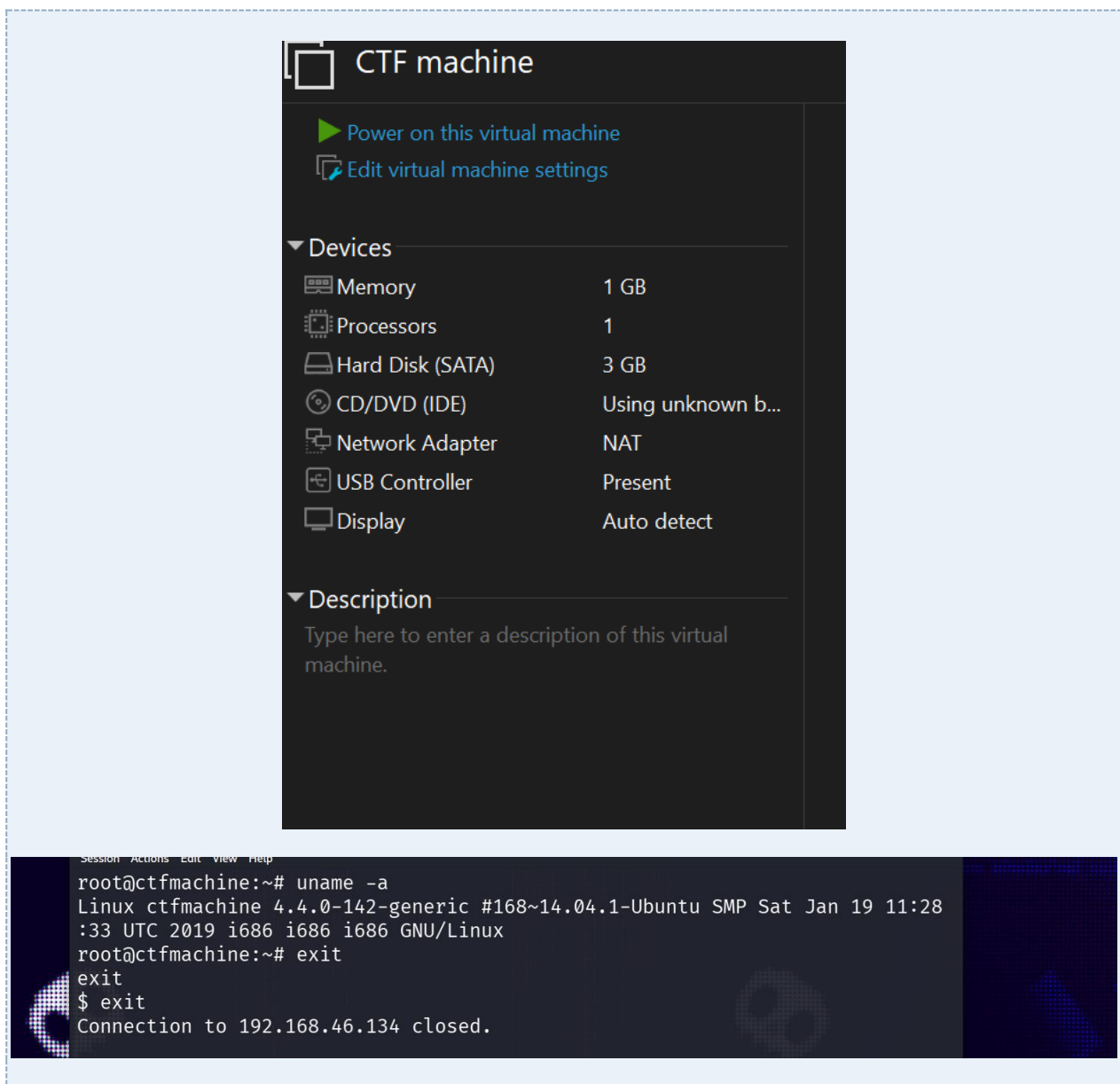
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2. Objective

The objective of this project is to design, configure, and deploy a Capture The Flag (CTF) machine that simulates a real-world penetration testing environment. The machine challenges participants to enumerate a web server, discover hidden credentials embedded within website content, escalate access via SSH, recover a hidden file using steganography, and capture two flags placed at the user and root levels of the filesystem.

This exercise demonstrates practical skills in Linux system administration, web hosting, firewall configuration, secure credential handling for educational purposes, and steganographic data hiding.

3. Machine Specifications



The image displays a virtual machine interface for a machine named "CTF machine". The interface includes a title bar with a window icon and the name "CTF machine". Below the title bar, there are two buttons: "Power on this virtual machine" (with a green play icon) and "Edit virtual machine settings" (with a wrench icon). The main content area is divided into two sections: "Devices" and "Description".

Devices

Device	Value
Memory	1 GB
Processors	1
Hard Disk (SATA)	3 GB
CD/DVD (IDE)	Using unknown b...
Network Adapter	NAT
USB Controller	Present
Display	Auto detect

Description

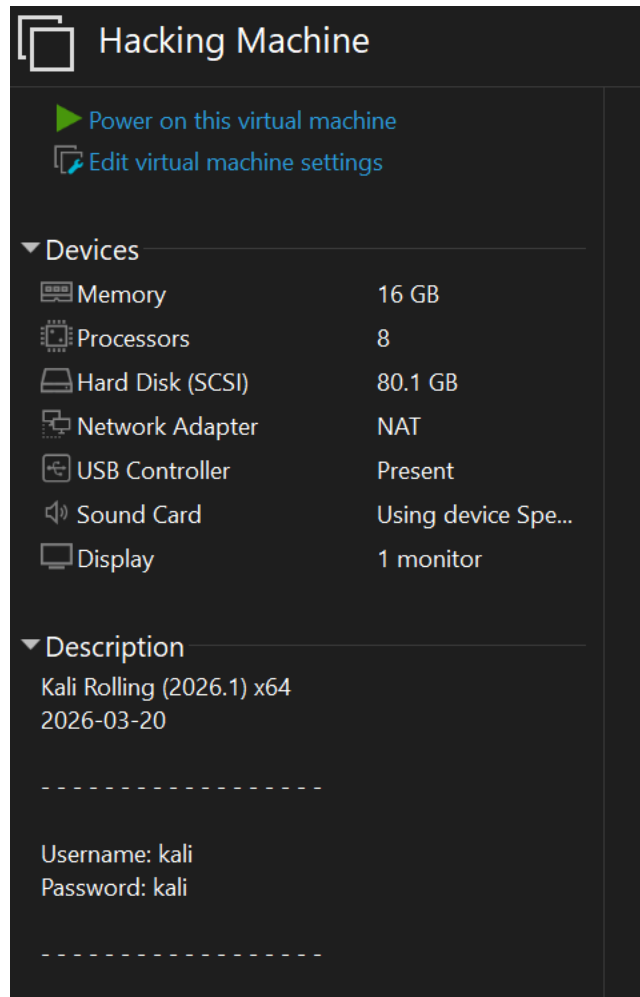
Type here to enter a description of this virtual machine.

Below the settings panel, there is a terminal window showing a session with the CTF machine. The terminal output is as follows:

```
root@ctfmachine:~# uname -a
Linux ctfmachine 4.4.0-142-generic #168~14.04.1-Ubuntu SMP Sat Jan 19 11:28
:33 UTC 2019 i686 i686 i686 GNU/Linux
root@ctfmachine:~# exit
exit
$ exit
Connection to 192.168.46.134 closed.
```

4. Operating System Used

State the distribution, version, and kernel release used for the CTF machine, and the rationale for choosing it (e.g., lightweight footprint, package availability, familiarity).



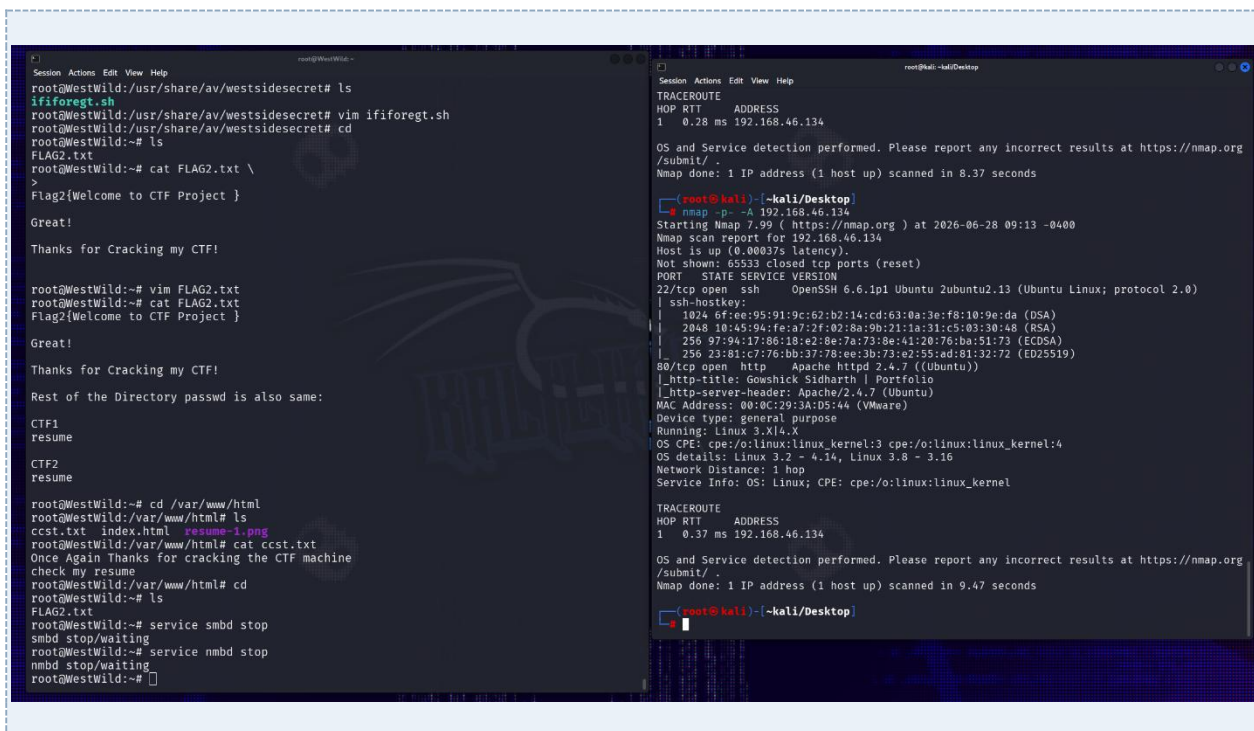
5. Network Configuration

Only two ports are exposed to participants:

Port	Service	Purpose
22	SSH	Remote shell access for credential-based login and privilege escalation
80	HTTP	Hosts the profile website containing enumeration clues

Firewall rules (e.g., via ufw or iptables) deny all other inbound traffic by default. Example policy:

- Default incoming policy: DENY
- Default outgoing policy: ALLOW
- Allow rule: 22/tcp (SSH)
- Allow rule: 80/tcp (HTTP)



```
root@WestWild: /usr/share/av/westsidesecret# ls
ififoregt.sh
root@WestWild: /usr/share/av/westsidesecret# vim ififoregt.sh
root@WestWild: /usr/share/av/westsidesecret# cd
root@WestWild: ~# ls
FLAG2.txt
root@WestWild: ~# cat FLAG2.txt \
>
Flag2{Welcome to CTF Project }

Great!

Thanks for Cracking my CTF!

root@WestWild: ~# vim FLAG2.txt
root@WestWild: ~# cat FLAG2.txt
Flag2{Welcome to CTF Project }

Great!

Thanks for Cracking my CTF!

Rest of the Directory passwd is also same:

CTF1
resume
CTF2
resume

root@WestWild: ~# cd /var/www/html
root@WestWild: /var/www/html# ls
ccst.txt index.html resume-1.png
root@WestWild: /var/www/html# cat ccst.txt
Once Again Thanks for cracking the CTF machine
check my resume
root@WestWild: /var/www/html# cd
root@WestWild: ~# ls
FLAG2.txt
root@WestWild: ~# service smbd stop
smbd stop/waiting
root@WestWild: ~# service nmbd stop
nmbd stop/waiting
root@WestWild: ~#
```

```
TRACEROUTE
HOP RTT ADDRESS
1 0.28 ms 192.168.46.134

OS and Service detection performed. Please report any incorrect results at https://nmap.org
/submit/.
Nmap done: 1 IP address (1 host up) scanned in 8.37 seconds

root@kali: ~# nmap -p -A 192.168.46.134
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-28 09:13 -0400
Nmap scan report for 192.168.46.134
Host is up (0.00037s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 1024 6f:ee:95:91:9c:62:b2:14:cd:63:0a:3e:f8:10:9e:da (DSA)
|_ 2048 18:45:94:fe:a7:2f:02:8a:9b:21:1a:31:c5:03:30:48 (RSA)
|_ 256 97:94:17:06:10:a2:0e:7a:73:8e:41:20:76:ba:51:73 (ECDSA)
|_ 256 23:81:c7:76:bb:37:78:ee:3b:73:62:55:ad:81:32:72 (ED25519)
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))
|_ http-title: Gowshick Sidharth | Portfolio
|_ http-server-header: Apache/2.4.7 (Ubuntu)
MAC Address: 00:0C:29:3A:C5:44 (VMware)
Device type: general purpose
Running: Linux 3.X14.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT ADDRESS
1 0.37 ms 192.168.46.134

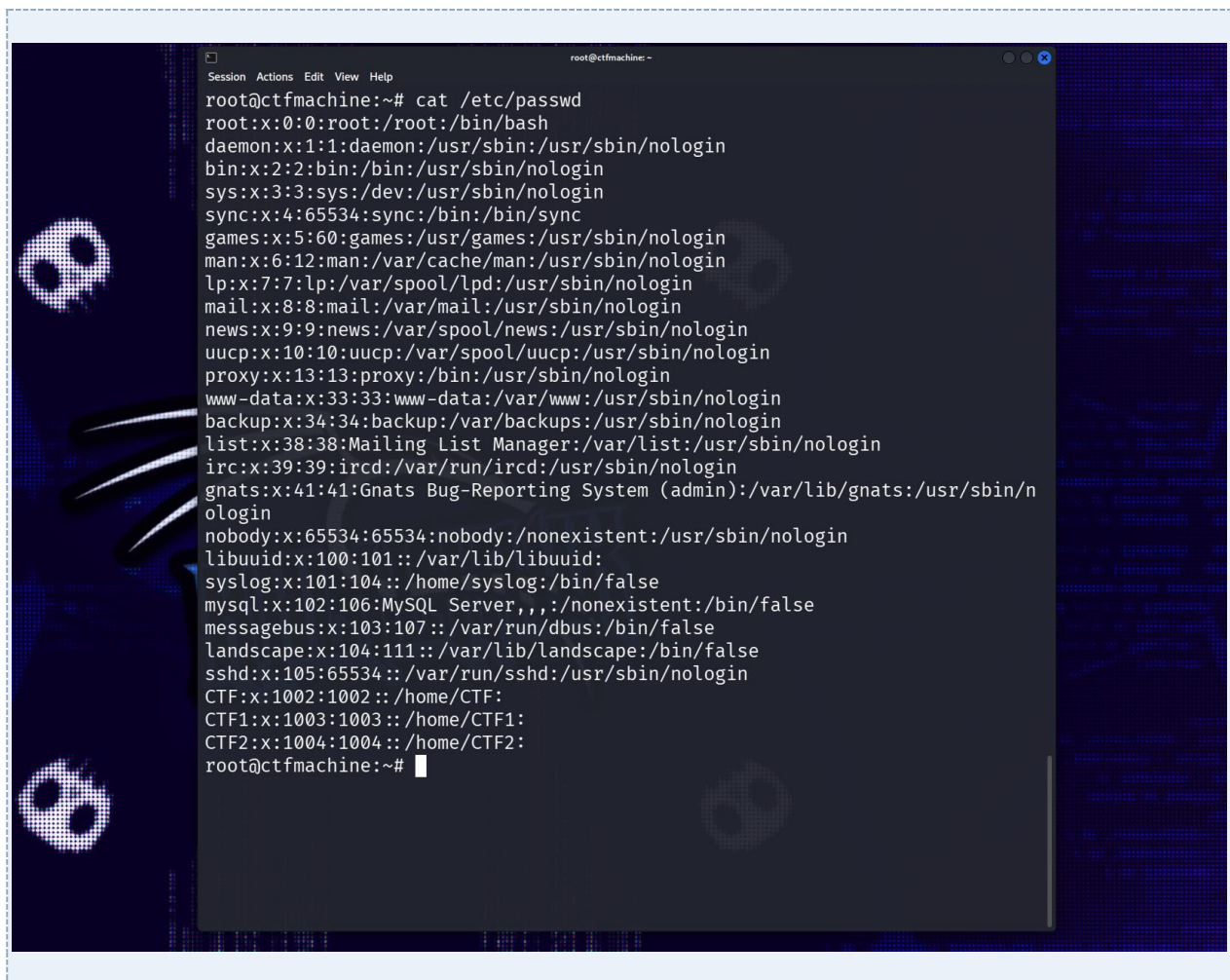
OS and Service detection performed. Please report any incorrect results at https://nmap.org
/submit/.
Nmap done: 1 IP address (1 host up) scanned in 9.47 seconds

root@kali: ~#
```

6. User Accounts Created

Three user accounts were created, each with its own home directory. Credentials are intentionally not disclosed in this report — they are discoverable only via enumeration of the website, as required by the assignment. The table below documents structure without revealing secrets.

Username	Home Directory	Role in Challenge
CTF	/home/	Initial low-privilege foothold
CTF1	/home/	Secondary account (lateral clue)
CTF2	/home/	Holds user.txt flag

A terminal window titled 'root@ctfmachine: ~' displays the output of the command 'cat /etc/passwd'. The output lists system and user accounts with their respective IDs, group IDs, names, home directories, and shell programs. The background of the terminal features a dark blue, pixelated skull graphic on the left side.

```
root@ctfmachine:~# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
libuuid:x:100:101::/var/lib/libuuid:
syslog:x:101:104::/home/syslog:/bin/false
mysql:x:102:106:MySQL Server,,,:/nonexistent:/bin/false
messagebus:x:103:107::/var/run/dbus:/bin/false
landscape:x:104:111::/var/lib/landscape:/bin/false
sshd:x:105:65534::/var/run/sshd:/usr/sbin/nologin
CTF:x:1002:1002::/home/CTF:
CTF1:x:1003:1003::/home/CTF1:
CTF2:x:1004:1004::/home/CTF2:
root@ctfmachine:~#
```

7. Services Configured

7.1 SSH Service

Describe the SSH server (e.g., OpenSSH), authentication method (password vs key), and any hardening applied or intentionally left permissive for the challenge.

7.2 Web Server

Describe the web server software (e.g., Apache2 or Nginx), document root path, and virtual host configuration used to serve the profile website on port 80

```

(root@kali)~[~kali/Desktop]
# nmap -A 192.168.46.134
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-30 13:29 -0400
Nmap scan report for 192.168.46.134
Host is up (0.00036s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
22/tcp    open  ssh            OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 6f:ee:95:91:9c:62:b2:14:cd:63:0a:3e:f8:10:9e:da (DSA)
|   2048 10:45:94:fe:a7:2f:02:8a:9b:21:1a:31:c5:03:30:48 (RSA)
|   256 97:94:17:86:18:e2:8e:7a:73:8e:41:20:76:ba:51:73 (ECDSA)
|_  256 23:81:c7:76:bb:37:78:ee:3b:73:e2:55:ad:81:32:72 (ED25519)
80/tcp    open  http           Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Gowshick Sidharth | Portfolio
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
MAC Address: 00:0C:29:3A:D5:44 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: Host: CTFMACHINE; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   NetBIOS computer name: CTFMACHINE\x00
|   Workgroup: WORKGROUP\x00
|_  System time: 2026-06-30T20:29:45+03:00
|_ nbstat: NetBIOS name: CTFMACHINE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ clock-skew: -2h59m59s
|_ smb2-time: Protocol negotiation failed (SMB2)

TRACEROUTE
HOP RTT      ADDRESS
1   0.36 ms  192.168.46.134

OS and Service detection performed. Please report any incorrect results at https://nmap.org/su
bmit/ .
Nmap done: 1 IP address (1 host up) scanned in 43.62 seconds

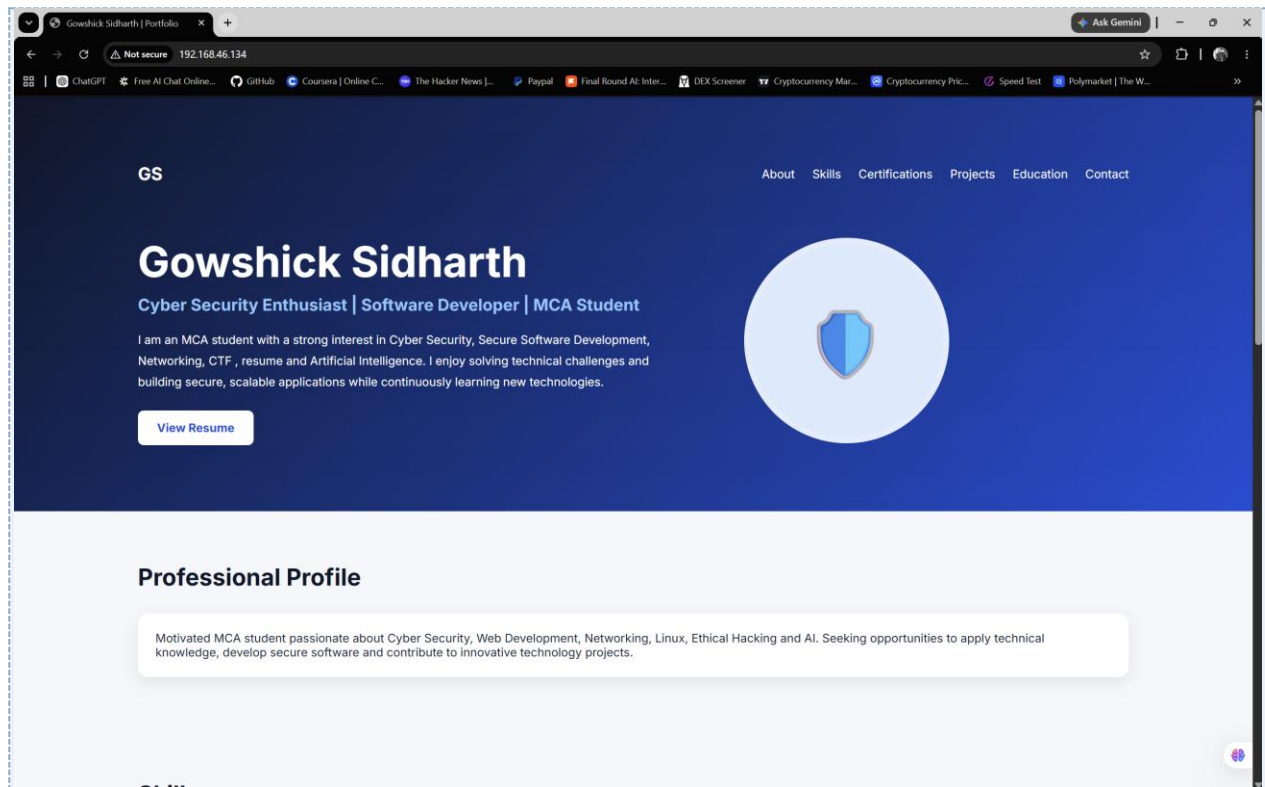
(root@kali)~[~kali/Desktop]
#

```

8. Website Design Overview

The hosted website acts as a personal profile site and the primary enumeration surface of the challenge. It includes the following sections:

- Professional profile / About Me
- Skills
- Certifications
- Projects
- Contact information
- Hidden hints embedded throughout the page structure



9. Hidden Credentials Methodology

Credentials are not stored in plaintext on the visible page. Instead, participants must enumerate the site using standard reconnaissance techniques. Document each technique used, for example:

Technique	Location
HTML comment	index.html
Hidden/unlinked page	View the resume by clicking the link in the website

CSS file comment	Css code is in the same index.html
Encoded string (Base64/ROT13)	user=CTF+pass=resume

10. Steganography Implementation

A resume document was embedded inside an image file placed on the Desktop using a steganography tool/technique. Document the tool, command, and verification method used.

11. Flag Locations

Flag	Path	Format
User Flag	/home/.txt	CTF{...}
Root Flag	/root/root.txt	CTF{...}

```
root@ctfmachine: -
Session Actions Edit View Help
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
libuuid:x:100:101::/var/lib/libuuid:
syslog:x:101:104::/home/syslog:/bin/false
mysql:x:102:106:MySQL Server,,,:/nonexistent:/bin/false
messagebus:x:103:107::/var/run/dbus:/bin/false
landscape:x:104:111::/var/lib/landscape:/bin/false
sshd:x:105:65534::/var/run/sshd:/usr/sbin/nologin
CTF:x:1002:1002::/home/CTF:
CTF1:x:1003:1003::/home/CTF1:
CTF2:x:1004:1004::/home/CTF2:
root@ctfmachine:~# ls
FLAG2.txt
root@ctfmachine:~# cat FLAG2.txt
Flag2{Welcome to CTF Project }

Great!

Thanks for Cracking my CTF!

Rest of the Directory passwd is also same:

CTF1
resume

CTF2
resume

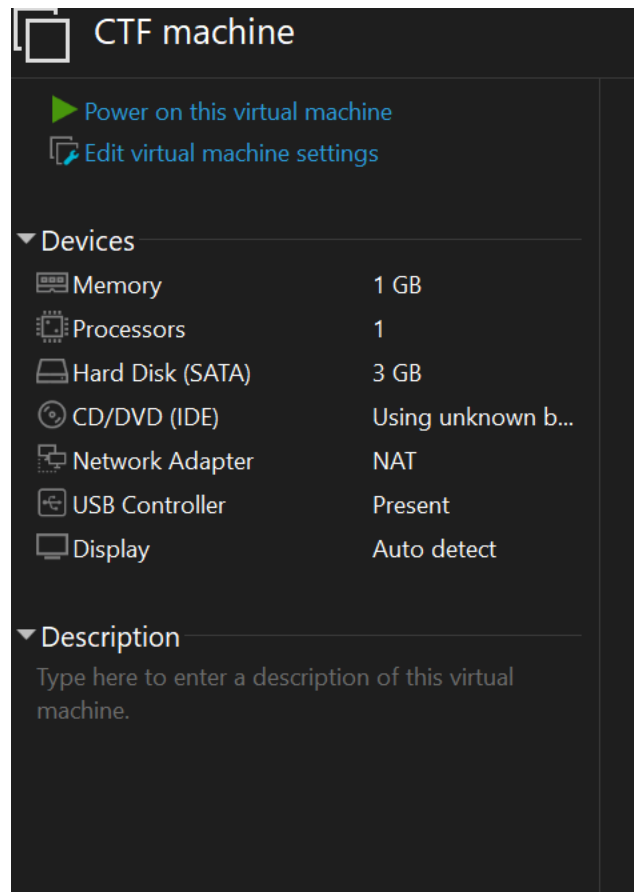
root@ctfmachine:~#
```

```
root@ctfmachine:/home/wavex/wave# cat FLAG1.txt
dXNlcj1DVEYrcGFzc21yZXN1bWUK
root@ctfmachine:/home/wavex/wave#
```

```
root@ctfmachine:/usr/share/av/websitesecret# ls
ififoregt.sh
root@ctfmachine:/usr/share/av/websitesecret# cat ififoregt.sh
#!/bin/bash
figlet "if i foregt so this my way"
echo "user:root"
echo "password:toor"

root@ctfmachine:/usr/share/av/websitesecret#
```

12. Screenshots of Configuration



```
Ubuntu 14.04.6 LTS ctfmachine tty1
192.168.46.134
ctfmachine login:
```

13. Testing Procedure

Describe the end-to-end test performed to validate the challenge is solvable as intended, e.g.:

- 1. Confirmed only ports 22 and 80 respond using an external port scan
- 2. Browsed the website and enumerated robots.txt, hidden pages, and source comments
- 3. Decoded the discovered credential string
- 4. Logged in via SSH using the recovered credentials
- 5. Located and read Flag1.txt
- 6. Performed privilege escalation to root
- 7. Located and read ifforegt.sh
- 8. Downloaded the Desktop image and extracted the hidden resume

```
(root@kali)~[~kali/Desktop]
# nmap -A 192.168.46.134
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-30 13:29 -0400
Nmap scan report for 192.168.46.134
Host is up (0.00036s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 6f:ee:95:91:9c:62:b2:14:cd:63:0a:3e:f8:10:9e:da (DSA)
|   2048 10:45:94:fe:a7:2f:02:8a:9b:21:1a:31:c5:03:30:48 (RSA)
|   256  97:94:17:86:18:e2:8e:7a:73:8e:41:20:76:ba:51:73 (ECDSA)
|_  256  23:81:c7:76:bb:37:78:ee:3b:73:e2:55:ad:81:32:72 (ED25519)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Gowshick Sidharth | Portfolio
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
MAC Address: 00:0C:29:3A:D5:44 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: Host: CTFMACHINE; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.3.11-Ubuntu)
|   NetBIOS computer name: CTFMACHINE\x00
|   Workgroup: WORKGROUP\x00
|_  System time: 2026-06-30T20:29:45+03:00
|_ nbstat: NetBIOS name: CTFMACHINE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ clock-skew: -2h59m59s
|_ smb2-time: Protocol negotiation failed (SMB2)

TRACEROUTE
HOP RTT      ADDRESS
1   0.36 ms  192.168.46.134

OS and Service detection performed. Please report any incorrect results at https://nmap.org/su
bmit/ .
Nmap done: 1 IP address (1 host up) scanned in 43.62 seconds

(root@kali)~[~kali/Desktop]
#
```

14. Vulnerability Walkthrough

Provide a step-by-step attacker-perspective walkthrough from initial enumeration to root flag capture. Include commands used at each stage and the reasoning behind each step.

```
(root@kali)~[/home/kali]
# ssh CTF@192.168.46.134
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
CTF@192.168.46.134's password:
Added user CTF.

Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)

 * Documentation:  https://help.ubuntu.com/
```

```

root@WestWild: ~
Session Actions Edit View Help
messagebus:*:18107:0:99999:7:::
landscape:*:18107:0:99999:7:::
sshd:*:18107:0:99999:7:::
CTF:$6$J1FxtBkb$DZU15rwlSvf08ewEgoovSBnib8SKZifkyA0RRKJqB6CmOG07yNk3B53N4JshLp7byIh4i7L
:7:::
CTF:$6$498Fjreu$e9B0u.TkZY8GvGyRV1ZgfmgSIYBji4jCQwvN/toyI27M82fv34lNvmCtdbHmv9VIR/vNXx
9:7:::
CTF2:$6$hygfmz1A$VkhZ2.3Krw8m8LsnUbIBrrzp5GT4ffcMvaLoCpInr0QVY3arhzX0EmJc0spjeGmT5o1xfXY
9:7:::
root@WestWild:~# hostnamectl set-hostname CTFMachine
root@WestWild:~# cd /home/wavex/wave
root@WestWild:~/home/wavex/wave# ls
FLAG1.txt message_from_aveng.txt
root@WestWild:~/home/wavex/wave# rm message_from_aveng.txt
root@WestWild:~/home/wavex/wave# ls
FLAG1.txt
root@WestWild:~/home/wavex/wave# cat FLAG1.txt
RmxhZzF7V2VsY29tZV9UMF9USEUtvZNTVC1XMUXELUIwcmRlcn0kdXNlcjps3YXZleApwYXNzd29yZDpkb29yK29
root@WestWild:~/home/wavex/wave# echo "user=CTF+pass=resume" | base64
dXNlcj1DVEYrcGFzc21yZXN1bWUK
root@WestWild:~/home/wavex/wave# vim FLAG1.txt
root@WestWild:~/home/wavex/wave# vim FLAG1.txt
root@WestWild:~/home/wavex/wave# cat FLAG1.txt
dXNlcj1DVEYrcGFzc21yZXN1bWUK
root@WestWild:~/home/wavex/wave# cd
root@WestWild:~# cd /usr/share/av/websitesecret
bash: cd: /usr/share/av/websitesecret: No such file or directory
root@WestWild:~# cd /usr/share/av/websitesecret
root@WestWild:~/usr/share/av/websitesecret# ls
ififoregt.sh
root@WestWild:~/usr/share/av/websitesecret# vim ififoregt.sh
root@WestWild:~/usr/share/av/websitesecret# cd
root@WestWild:~# ls
FLAG2.txt
root@WestWild:~# cat FLAG2.txt \
>
Flag2{Welcome to CTF Project }

Great!

Thanks for Cracking my CTF!

root@WestWild:~# vim FLAG2.txt
root@WestWild:~# cat FLAG2.txt

```

15. Conclusion

Summarize what the exercise demonstrated, skills practiced (web enumeration, Linux administration, firewall hardening, steganography, privilege escalation), and any lessons learned or potential improvements for future iterations of the machine.